DS605: Fundamentals of Machine Learning

Lab Assignment - 6

Feature Extraction and Machine Learning with Image and Text Data

Image Dataset: Asphalt Crack Dataset - 400 Images (Mendeley Data)

Text Dataset: Email Spam Classification Dataset - 5,172 Emails (Kaggle)

Submission: Public GitHub repository link only.

Objective

Convert raw image and text data into numerical feature representations and use traditional machine-learning models for classification.

Part A - Image Feature Extraction and Classification

- Read the asphalt images using OpenCV or PIL, inspect their dimensions/channels, resize them to a common size, and create grayscale representations.
- Using NumPy, extract numerical features such as mean brightness, contrast, dark-pixel ratio, bright-pixel ratio, and related intensity statistics.
- Use OpenCV Canny edge detection to compute edge-based features such as edge count or edge density. Create one feature row per image with the crack/non-crack label.
- Train suitable traditional classifiers and evaluate using accuracy, precision, recall, F1-score, confusion matrix, training time, and prediction time.

Part B - Text Vectorization and Spam Classification

- Load the email dataset, inspect the spam/non-spam distribution, and perform suitable basic text cleaning.
- Convert email text into numerical vectors using CountVectorizer and TF-IDF Vectorizer, then train suitable classification models on both representations.
- Compare predictive performance, number of generated features, training time, and prediction time for the two text representations.

Part C - Improve the Representation

- For either task, make at least one justified change to improve accuracy or reduce computation, such as changing image features/Canny thresholds, normalization, limiting vocabulary, stop-word handling, n-grams, or model choice.
- Compare the original and improved approaches and discuss the trade-off between feature dimensionality, computation time, and classification performance.

Expected Workflow

Raw Image/Text -> Preprocessing -> Feature Extraction / Vectorization -> Train-Test Split -> ML Model -> Evaluation -> Representation Improvement

Important Requirement

- Do not use CNNs, deep-learning models, or pretrained image embeddings. Image features must be derived from the raw images using NumPy/OpenCV/PIL; Scikit-learn may be used for vectorizers and traditional ML models.

Deliverables (GitHub Only)

- Public repository containing runnable code/notebook, sample image and Canny-edge visualizations, extracted image-feature table, text-vectorization comparison, evaluation results, and a concise README with observations.
- Submit only the public GitHub repository link.